Physical Chemistry Cume
Chemical Kinetics

The addition of hydrogen halides to alkenes has played a fundamental role in the investigation of organic reaction mechanisms. In one study, hydrogen chloride and propene were examined over a range of temperatures and the amount of 2-chloropropane formed was determined by NMR.

$$HCl + CH_3CH{=}CH_2 \longrightarrow CH_3CHClCH_3$$

(a) Show that, if the reaction A+B → P proceeds for a short time δt, the concentration of product follows $\frac{[P]}{[A]} = k_r[A]^{m-1}[B]^n \delta t$, if the reaction is *m*th-order in A and *n*th-order in B. (25 pts)

(b) In a series of runs the ratio of [chloropropane] to [propene] was independent of [propene], but the ratio of [chloropropane] to [HCl] for constant amounts of propene was proportional to $[HCl]^2$. What are the orders of the reaction with respect to each reactant? Write the rate law of the reaction in terms of formation of the product. (25 pts)

(c) Show that the following mechanism can account for the rate law of the reaction (25 pts):

$$HCl + HCl \rightleftharpoons (HCl)_2 \qquad K_1$$

$$HCl + CH_3CH{=}CH_2 \rightleftharpoons \text{complex} \qquad K_2$$

$$(HCl)_2 + \text{complex} \longrightarrow CH_3CHClCH_3 + 2\,HCl \qquad k_r \text{ (slow)}$$

(d) In the experiment, an inverse temperature dependence of the reaction rate was observed, the overall rate of reaction at 70 °C being approximately one-third that at 19 °C. Estimate the apparent activation energy and the activation energy of the rate-determining step given that the enthalpies of the two equilibra (K_1 and K_2) are both of the order of -14 $kJmol^{-1}$. (25 pts)